

CSA05-DATABASE MANAGEMENT SYSTEMS

LAB OBSERVATION

EXPERIMENT:02

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands.

QUESTION:

- 1) Alter the table STUDENT2 with following structure.

#	Column Name	Constraints
1	RegNo	PRIMARY KEY

SQL QUERIES:

```
mysql> ALTER TABLE STUDENT2
-> ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.03 sec)
```

OUTPUT:

```
mysql> DESC STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int           | NO   | PRI | NULL    |       |
| Name  | varchar(30)   | YES  |     | NULL    |       |
| DeptNo | int           | YES  |     | NULL    |       |
| Age   | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

- 2) Alter the table name FACULTY with following structure.

#	Column Name	Constraints
1	FacNo	PRIMARY KEY
2	Gender	CHECK 'M' or 'F'

SQL QUERIES:

```
mysql> ALTER TABLE FACULTY
-> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.03 sec)
```

OUTPUT:

```
mysql> DESC FACULTY;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo      | int           | NO   | PRI | NULL    |       |
| FacName    | varchar(30)   | YES  |     | NULL    |       |
| Gender     | char(1)       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

3) Alter the table name DEPARTMENT with following structure.

#	Column Name	Constraints
1	DeptNo	PRIMARY KEY

SQL QUERIES:

```
mysql> ALTER TABLE DEPARTMENT
-> ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.03 sec)
```

OUTPUT:

```
mysql> DESC DEPARTMENT;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo     | int           | NO   | PRI | NULL    |       |
| DeptName   | varchar(30)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

4) Alter the table name COURSE with following structure.

#	Column Name	Constraints
1	CourseNo	PRIMARY KEY
2	SemNo	1 to 6

SQL QUERIES:

```
mysql> ALTER TABLE COURSE
-> ADD PRIMARY KEY (CourseNo);
Query OK, 0 rows affected (0.03 sec)
```

OUTPUT:

```
mysql> DESC COURSE;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo   | int           | NO   | PRI | NULL    |       |
| CourseName | varchar(40)   | YES  |     | NULL    |       |
| SemNo      | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

5) Alter the table name STUDENT2 with following structure.

#	Column Name	Constraints
1	S1Name	UNIQUE KEY

SQL QUERIES:

```
mysql> ALTER TABLE STUDENT2
-> ADD CONSTRAINT uq_s1name
-> UNIQUE (S1Name);
Query OK, 0 rows affected (0.03 sec)
```

OUTPUT:

```
mysql> DESC STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo      | int           | NO   | PRI | NULL    |       |
| S1Name     | varchar(30)   | YES  | UNI | NULL    |       |
| DeptNo     | int           | YES  |     | NULL    |       |
| Age        | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

6) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table.

SQL QUERIES:

```
mysql> ALTER TABLE STUDENT3
-> ADD CONSTRAINT fk_regno
-> FOREIGN KEY (RegNo)
-> REFERENCES STUDENT2(RegNo);
Query OK, 0 rows affected (0.04 sec)
```

OUTPUT:

```
mysql> DESC STUDENT3;
```

Field	Type	Null	Key	Default	Extra
StuID	int	NO	PRI	NULL	
RegNo	int	YES	MUL	NULL	
Address	varchar(50)	YES		NULL	

3 rows in set (0.00 sec)

RESULT:

DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.